PROBABILITY BASICS

$0 \le P(A) \le 1, \quad P(A) + P(A') = 1$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B \cup C) = \sum P - \sum P(\text{pairs}) + P(A \cap B \cap C)$

Mut. Excl: $A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$

Conditional & Independence

$P(B|A) = \frac{P(A \cap B)}{P(A)}$

Indep: $P(A \cap B) = P(A)P(B)$

Bayes' / Total Probability

$P(A) = \sum_i P(A|B_i)P(B_i)$

$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$

DISCRETE VS CONTINUOUS

Discrete: counting, PMF, $\sum$. **Continuous:** measuring, PDF, $\int$.

	Disc	Cont
Prob fn	PMF $f(x)$	PDF $f(x)$
CDF	$\sum_{t \le x} f(t)$	$\int_{-\infty}^{x} f(t)\,dt$
$E[X]$	$\sum x f(x)$	$\int x f(x)\,dx$
$P(X=a)$	$f(a)$	$=0$
$P(X>a)$	$\sum_{x>a} f(x)$	$\int_a^{\infty} f(x)\,dx$

CDF & PDF

$P(a < X \le b) = F(b) - F(a)$

CDF check: non-decr, $0 \le F \le 1$, $F(-\infty)=0$, $F(\infty)=1$

PDF check: $f(x) \ge 0$ and $\int f = 1$. Link: $f = F'$

JOINT & MARGINALS

Valid: $f \ge 0$, total$=1$. $g(x) = \sum_y f(x,y)$, $h(y) = \sum_x f(x,y)$

Cont: $g(x) = \int f(x,y)\,dy$, $h(y) = \int f(x,y)\,dx$

Find c: set $\sum\sum f = 1$ or $\iint f = 1$, solve.

Conditional Distribution

$f(x|y) = \frac{f(x,y)}{f(y)}$

$P(a \le X \le b \mid c \le Y \le d) = \frac{\int_c^d \int_a^b f\,dx\,dy}{P(c \le Y \le d)}$

Independence

$f(x,y) = g(x) \cdot h(y)\ \forall (x,y) \iff$ indep

One failure $\Rightarrow$ not independent!

Also: is support a Cartesian product? If no $\Rightarrow$ dependent.

Ex: Uniform on circle: support not a rect $\Rightarrow$ not indep even if Cov $= 0$.

E[X], VAR, COV

$E[X] = \sum x\, g(x) = \sum_x \sum_y x\, f(x,y)$

Cont: $E[X] = \iint x\, f(x,y)\,dy\,dx$

$E[XY] = \sum_x \sum_y xy\, f(x,y)$

$E[aX+b] = aE[X] + b$; $\quad E[X+Y] = E[X] + E[Y]$

$E[XY] = E[X]E[Y]$ **only if indep!**

μ (**mean**) $= E[X]$. "Avg cost"$=E[X]\times$unit.

Variance

σ^2 (**variance**) $= \mathrm{Var}(X) = E[X^2] - (E[X])^2$

σ (**std dev**) $= \sqrt{\mathrm{Var}(X)}$

$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$

$\mathrm{Var}(aX+bY) = a^2V_X + b^2V_Y + 2ab\,\mathrm{Cov}$

Indep: $\mathrm{Var}(X+Y) = V_X + V_Y$

Covariance

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$

$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; symmetric

$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$

Indep $\Rightarrow$ Cov $= 0$. **Cov $= 0 \not\Rightarrow$ indep!**

$\rho = \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$

CH5: DISCRETE DISTRIBUTIONS

When to Use What

Situation	Distribution
Fixed n trials, # succ	Binomial
n trials, k categories	Multinomial
Draw w/o replacement	Hypergeometric
# trials until k-th succ	Neg. Binomial
# trials until 1st succ	Geometric
# events in interval	Poisson

Binomial $X \sim \mathrm{Bin}(n,p)$

n=# trials, p=succ prob, $q=1-p$, x=# successes.

$f(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, \dots, n$

$\mu = np, \quad \sigma^2 = npq$

"Exactly x": $\binom{n}{x} p^x q^{n-x}$

"Specific sequence": $p^x q^{n-x}$ (no $\binom{n}{x}$!)

Multinomial

k categories, $\sum p_i = 1$, n trials.

$f = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}$

Hypergeometric

N=pop size, K=# successes (defectives) in pop, n=sample size, x=# successes drawn. **W/o replacement.**

$P(X=x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$

$\mu = \frac{nK}{N}, \quad \sigma^2 = n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$

If $n/N < 0.05$: approx with $\mathrm{Bin}(n,\ p=K/N)$

Acceptance Sampling

Lot N items, K defective, sample n, ship if $X \le c$.

$P(\text{ship}) = \sum_{x=0}^{c} \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$

Ex: Box 25, $K=3$ def, sample $n=3$, ship if 0 def.

W/o replacement: $P(ship) = \frac{\binom{3}{0}\binom{22}{3}}{\binom{25}{3}}$

W/ replacement $p = K/N = 3/25$.

$P(ship) = (1 - \frac{3}{25})^3$ This is Binomial!

Neg. Binomial $X \sim \mathrm{NB}(k,p)$

Use when: # trials until k-th success.

x (**= total # trials**), k=target successes, p=succ prob.

$P(X=x) = \binom{x-1}{k-1} p^k q^{x-k}, \quad x = k, k+1, \dots$

$\mu = k/p, \quad \sigma^2 = kq/p^2$

Ex: Detector $p = 0.05$. 3rd activation on 12th: $k=3$, $x=12$. $\binom{11}{2}(0.05)^3(0.95)^9$

Geometric $\mathrm{Geo}(p)$

Use when: # trials until 1st success.

NB with $k=1$. $\quad P(X=x) = pq^{x-1}$, $x = 1, 2, \dots$

$\mu = 1/p, \quad \sigma^2 = q/p^2$

Memoryless: $P(X > s+t \mid X > s) = P(X > t)$

Poisson $\mathrm{Poi}(\lambda t)$

Use when: # events in a time/space interval.

λ=rate (avg per unit time), t=interval length.

$P(X=x) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots$

$\mu = \lambda t, \quad \sigma^2 = \lambda t \quad (\mu = \sigma^2!)$

Sum of indep Poissons: $\mathrm{Poi}(\mu_1 + \mu_2)$

Poisson Approx to Binomial

n large, p small: $\mathrm{Bin}(n,p) \approx \mathrm{Poi}(\mu = np)$

$n \ge 20$, $p \le 0.05$; or $n \ge 100$, $np \le 10$.

CH6: CONTINUOUS DISTRIBUTIONS

Uniform $U(A,B)$

$f(x) = \frac{1}{B-A}, \ A \le x \le B$

$\mu = \frac{A+B}{2}, \quad \sigma^2 = \frac{(B-A)^2}{12}$

$P(c \le X \le d) = \frac{d-c}{B-A}$ (for $A \le c \le d \le B$)

Normal $N(\mu, \sigma^2)$

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$

Standardize: $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$

$X = \mu + Z\sigma$

Z-Table Rules

$P(Z < z)$ read directly from table

$P(Z > z) = 1 - P(Z < z)$

$P(Z > z) = P(Z < -z)$ (symmetry)

$P(a < Z < b) = P(Z < b) - P(Z < a)$

For $X \sim N(\mu, \sigma^2)$, $a > 0$:

$P(X > a) = 1 - P\left(Z < \frac{a-\mu}{\sigma}\right)$

$P(X < a) = P\left(Z < \frac{a-\mu}{\sigma}\right)$

68-95-99.7 Rule

$P(\mu \pm 1\sigma) = 0.6826, \quad P(\mu \pm 2\sigma) = 0.9544$

$P(\mu \pm 3\sigma) = 0.9974$

Normal Approx to Binomial

$\mathrm{Bin}(n,p) \approx N(np = \mu,\ npq = \sigma^2)$ for large n.

Discrete $\to$ continuous rule:

$P(X \ge k) = 1 - P(X \le k-1)$ (integers only)

Continuity correction:

$P(X \le k) \approx \Phi\left(\frac{k+0.5-np}{\sqrt{npq}}\right)$

$P(X = k) \approx P(k-0.5 < Y < k+0.5)$

Normal Approx to Poisson

$\mathrm{Poi}(\lambda) \approx N(\lambda, \lambda)$ when λ is large ($\lambda > 5$).

$Z = \frac{X-\lambda}{\sqrt{\lambda}}$, with continuity correction.

Gamma $\mathrm{Gamma}(\alpha, \beta)$

α = shape (# events / arrivals), β = scale (mean time between events $= 1/\lambda$).

$f(x) = \frac{x^{\alpha-1}e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, \ x > 0$

$\mu = \alpha\beta, \quad \sigma^2 = \alpha\beta^2$

$\Gamma(n) = (n-1)!$; $\quad \Gamma(1/2) = \sqrt{\pi}$; $\quad \Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$

Exponential $\mathrm{Exp}(\beta)$

Gamma with $\alpha = 1$. **Time between Poisson events.**

β = mean time between events $= 1/\lambda$, where λ = Poisson rate (# events/unit time).

$f(x) = \frac{1}{\beta}e^{-x/\beta}, \quad F(x) = 1 - e^{-x/\beta}$

$\mu = \beta$, $\sigma^2 = \beta^2$. $\quad P(X > t) = e^{-t/\beta}$

Poisson Process Links

arrivals in t: $\mathrm{Poi}(\lambda t)$

Time to 1st: $\mathrm{Exp}(1/\lambda)$

Time to α-th: $\mathrm{Gamma}(\alpha, 1/\lambda)$

Sum of n indep $\mathrm{Exp}(\beta)$=$\mathrm{Gamma}(n,\beta)$

Weibull

$f(x)(pdf) = \alpha\beta x^{\beta-1}e^{-\alpha x^\beta}, \ x > 0$

$F(x)(CDF) = 1 - e^{-\alpha x^\beta}$

$\mu = \alpha^{-1/\beta}\Gamma(1 + 1/\beta)$

$\sigma^2 = \alpha^{-2/\beta}\left\{\Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2\right\}$

Hazard Rate

$R(t) = 1 - F(t)$ (reliability), $\quad z(t) = \frac{f(t)}{1-F(t)}$

Weibull: $z(t) = \alpha\beta t^{\beta-1}$

$\beta = 1$: constant; $\beta > 1$: wear-out; $\beta < 1$: strengthen

Chi-Squared $\chi^2(\nu)$

$\mathrm{Gamma}(\nu/2, 2)$. $\quad \mu = \nu$, $\sigma^2 = 2\nu$

$Z_1^2 + \cdots + Z_\nu^2 \sim \chi^2(\nu)$

CH7: TRANSFORMATIONS

Discrete: 1-to-1

$Y = u(X)$, inverse $x = w(y)$: $g(y) = f[w(y)]$

Discrete: Not 1-to-1

$g(y) = \sum_{\{x:\, u(x)=y\}} f(x)$

Ex: $Y = X^2$, $X \in \{-2, \dots, 2\}$: $P(Y=4) = P(X=2) + P(X=-2)$

CDF Method (continuous)

$F_Y(y) = P(u(X) \le y)$, then $f_Y = F_Y'$.

Constant range ($X > 0$, $Y = X^2$): $F_Y = P(X \le \sqrt{y}) = F_X(\sqrt{y})$.

$F_Y = P(-\sqrt{y} \le X \le \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} f(x)\,dx$.

Ex: $f(x) = (x+1)/2$, $-1 < x < 1$, $Y = X^2$: $F_Y = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{x+1}{2}\,dx = \sqrt{y} \Rightarrow f_Y = \frac{1}{2\sqrt{y}}$.

Jacobian Method (1-to-1)

$g(y) = f[w(y)] \cdot |J|, \quad J = w'(y) = \frac{dx}{dy}$

Continuous: Non-1-to-1

$g(y) = f[w_1(y)]|J_1| + f[w_2(y)]|J_2| + \cdots$

Ex: Same ex above via Jacobian: $w_1 = \sqrt{y}$, $w_2 = -\sqrt{y}$, $|J| = \frac{1}{2\sqrt{y}}$:

$g(y) = f(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + f(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}$.

Joint Transform (2 vars)

$g(y_1, y_2) = f[w_1, w_2] \cdot |J|$

$J = \frac{\partial x_1}{\partial y_1}\frac{\partial x_2}{\partial y_2} - \frac{\partial x_1}{\partial y_2}\frac{\partial x_2}{\partial y_1}$

Define Y_2 to make invertible, then marginalize.

Key Results

Sum indep $\mathrm{Exp}(\beta) \to \mathrm{Gamma}(n,\beta)$

Sum indep Poi$\to$Poi(sum means)

Sum indep $N \to N$(sum μ, sum σ^2)

$\left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$ if $X \sim N(\mu, \sigma^2)$

CH7: MOMENTS & MGFS

Moments

$\mu_r' = E[X^r]$ (r-th moment about zero)

Central moments: $E[(X-\mu)^r]$: 2nd$=\sigma^2$ (var), 3rd=skewness, 4th=kurtosis

MGF

$M_X(t) = E(e^{tX}) = \begin{cases} \sum_x e^{tx} f(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx, & X \text{ continuous} \end{cases}$

$\frac{d^r}{dt^r} M_X(t)\Big|_{t=0} = E[X^r]$:

$E[X] = M_X'(0), \quad E[X^2] = M_X''(0)$

$\mathrm{Var} = M_X''(0) - [M_X'(0)]^2$

MGF Properties

Linear: $Y = a + bX$: $M_Y(t) = e^{at}M_X(bt)$

Sum of indep: $Y = X_1 + \cdots + X_n$: $M_Y(t) = M_{X_1}(t) \cdot M_{X_2}(t) \cdots M_{X_n}(t)$

Common MGFs

Dist	MGF
$N(\mu, \sigma^2)$	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$
$\chi^2(\nu)$	$(1-2t)^{-\nu/2}$
$\mathrm{Poi}(\lambda)$	$e^{\lambda(e^t - 1)}$
$\mathrm{Exp}(\beta)$	$(1-\beta t)^{-1}$, $t < 1/\beta$
$\mathrm{Gamma}(\alpha, \beta)$	$(1-\beta t)^{-\alpha}$, $t < 1/\beta$

USEFUL SUMS & SERIES

$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \sum_{i=1}^n a = na$

$\sum_{i=0}^n x^i = \frac{1-x^{n+1}}{1-x}, \ x \ne 1$ (geometric)

$\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}, \ |x| < 1 \quad \sum_{i=1}^{\infty} x^i = \frac{x}{1-x}$

$\sum_{i=1}^{\infty} ix^{i-1} = \frac{1}{(1-x)^2}, \ |x| < 1$

$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ (Taylor series)

CH7: LIN. COMB. OF NORMALS

$X_i \sim N(\mu_i, \sigma_i^2)$ indep $\Rightarrow$

$Y = \sum a_i X_i \sim N\left(\sum a_i\mu_i,\ \sum a_i^2\sigma_i^2\right)$

Sum: $\sum X_i \sim N\left(\sum \mu_i, \sum \sigma_i^2\right)$

IID: $X_i \sim N(\mu, \sigma^2)$:

$\sum X_i \sim N(n\mu, n\sigma^2)$; $\quad \bar{X} \sim N(\mu, \sigma^2/n)$

Chi-Squared Additivity

$X_i \sim \chi^2(\nu_i)$ indep $\Rightarrow \sum X_i \sim \chi^2(\sum \nu_i)$

Cor: $X_i \sim N(\mu_i, \sigma_i^2)$ indep (or iid with same μ, σ): $Y = \sum\left(\frac{X_i - \mu_i}{\sigma_i}\right)^2 \sim \chi^2(n)$

CH8: SAMPLING DISTRIBUTIONS

Sampling Basics

$\bar{X} = \frac{1}{n}\sum X_i, \quad S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$

σ^2 = **population** variance (parameter, fixed, usually unknown)

S^2 = **sample** variance (statistic, random, estimate of σ^2)

σ given $\Rightarrow$ use Z. $\quad s$ given $\Rightarrow$ use t.

Normal Sampling & CLT

$X_i \sim N(\mu, \sigma^2)$: $\bar{X} \sim N(\mu, \sigma^2/n)$ **exactly**

CLT: any dist, $n \ge 30$: $\bar{X} \overset{\text{approx}}{\sim} N(\mu, \sigma^2/n)$

$Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ (σ known)

$\sum X_i \sim N(n\mu, n\sigma^2)$

Two-sample: $\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$

Chebyshev & Markov

Markov ($X \ge 0$, only need $E[X]$): $P(X \ge a) \le \frac{E[X]}{a}$

Chebyshev (need μ and σ^2, **any** dist): $P(\mu - k\sigma < X < \mu + k\sigma) \ge 1 - \frac{1}{k^2}$

Equiv: $P(|X-\mu| \ge \varepsilon) \le \frac{\sigma^2}{\varepsilon^2}$ (set $\varepsilon = k\sigma$)

WLLN proof: apply Cheby to $\bar{X}_n$:

$P(|\bar{X}_n - \mu| \ge \varepsilon) \le \frac{\sigma^2}{n\varepsilon^2} \to 0$

χ^2, Independence of $\bar{X}$ & S^2

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$ (normal pop)

$\left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$

Additivity: $\chi^2(\nu_1)+\chi^2(\nu_2)\sim\chi^2(\nu_1+\nu_2)$

$\bar{X}$ and S^2 are independent (normal pop only!)

t-Distribution

$T=\frac{Z}{\sqrt{V/\nu}}\sim t(\nu)$, $Z\sim N(0,1)$, $V\sim\chi^2(\nu)$ indep

One-sample: $T=\frac{\bar{X}-\mu}{S/\sqrt{n}}\sim t(n-1)$

$E(T)=0$, $\mathrm{Var}(T)=\frac{\nu}{\nu-2}$ $(\nu>2)$

As $\nu\to\infty$: $t(\nu)\to N(0,1)$

Reading table: $P(T>t_\alpha)=\alpha$

Ex: $t_{0.025}$ w/ $\nu=15$: look row 15, col 0.025 = 2.131

Scenario	Stat	Dist
σ known	$Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$	$N(0,1)$
σ unknown	$T=\frac{\bar{X}-\mu}{S/\sqrt{n}}$	$t(n-1)$
$n\geq 30$	either	$\approx N(0,1)$

F-Distribution

$F=\frac{U/\nu_1}{V/\nu_2}\sim F(\nu_1,\nu_2)$

Compare variances: $F=\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F(n_1-1,n_2-1)$

Under H_0: $\sigma_1^2=\sigma_2^2$: $F=\frac{S_1^2}{S_2^2}$

$F_{1-\alpha}(\nu_1,\nu_2)=\frac{1}{F_\alpha(\nu_2,\nu_1)}$ (reciprocal trick)

Hypothesis Testing Steps

1. State H_0, H_a 2. Test statistic
3. Rejection region or p-value
4. Decision 5. Conclusion in context

CI includes $\mu_0\Leftrightarrow$ fail to reject H_0

CH9: ESTIMATION

Point Estimation

Unbiased: $E(\hat{\Theta})=\theta$

$E(\bar{X})=\mu$, $E(S^2)=\sigma^2$ (why $n-1$)

Efficient: smallest variance among unbiased

Std Error: $\mathrm{SE}(\bar{X})=\frac{\sigma}{\sqrt{n}}$

Std dev = pop spread; Std error = estimator spread

Confidence Intervals

$\bar{X}\pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ (σ known), $\bar{X}\pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$ (σ unknown)

Width $\uparrow$ as confidence$\uparrow$, $\sigma\uparrow$, $n\downarrow$

How to Read Tables (Simple)

Z table: usually gives left-tail area $\Phi(z)=P(Z\leq z)$.

t table: entry $t_{\alpha,\nu}$ means $P(T_\nu>t_{\alpha,\nu})=\alpha$ (right tail).

χ^2 table: entry $\chi^2_{\alpha,\nu}$ means $P(\chi^2_\nu>\chi^2_{\alpha,\nu})=\alpha$.

F table: entry F_{α,ν_1,ν_2} means $P(F_{\nu_1,\nu_2}>F_{\alpha,\nu_1,\nu_2})=\alpha$.

CH9: CONFIDENCE INTERVALS

Margin of Error(E) + Generic CI

$$E=z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\quad\text{(if }\sigma\text{ known)}$$

$$E=t_{\alpha/2,\,n-1}\frac{S}{\sqrt{n}}\quad\text{(if }\sigma\text{ unknown)}$$

CI for μ: $(\bar{X}-E,\ \bar{X}+E)=\bar{X}\pm E$

Common z Values

Two-sided CIs ($z_{\alpha/2}$): 90% $\to$ 1.645, 95% $\to$ 1.96, 99% $\to$ 2.576.

One-sided bounds (z_α): 90% $\to$ 1.282, 95% $\to$ 1.645, 99% $\to$ 2.326.

CI for μ (σ known)

$\bar{X}\pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

CI for μ (σ unknown)

$\bar{X}\pm t_{\alpha/2,\,n-1}\frac{S}{\sqrt{n}}$ (need normal pop)

σ known $\Rightarrow Z$. σ unknown $\Rightarrow t$.

Sample Size for Error e

$n\geq\left(\frac{z_{\alpha/2}\,\sigma}{e}\right)^2$ (round UP)

Ex: $\sigma=40$, $e=10$, 95%: $n\geq(1.96\times 40/10)^2=61.5\Rightarrow 62$

One-Sided CI (σ known)

Upper: $\mu\leq\bar{X}+z_\alpha\frac{\sigma}{\sqrt{n}}$

Lower: $\mu\geq\bar{X}-z_\alpha\frac{\sigma}{\sqrt{n}}$

Use z_α not $z_{\alpha/2}$! All α in one tail.

CI for Proportion p

$\hat{p}=\frac{x}{n}$, where x = number of successes, n = sample size.

$\hat{p}\pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$, $\hat{q}=1-\hat{p}$

Need $n\hat{p}\geq 5$, $n\hat{q}\geq 5$.

Use z (not t): large n needed anyway, so $t\approx z$.

Prediction Interval

σ known: $\bar{X}\pm z_{\alpha/2}\,\sigma\sqrt{1+\frac{1}{n}}$

σ unknown: $\bar{X}\pm t_{\alpha/2,n-1}\,S\sqrt{1+\frac{1}{n}}$

PI > CI always! Reason:

$$\mathrm{Var}(\bar{X})=\frac{\sigma^2}{n}\quad\text{(mean-estimation uncertainty only)}$$

$$\mathrm{Var}(X_{\text{new}}-\bar{X})=\sigma^2+\frac{\sigma^2}{n}=\sigma^2\left(1+\frac{1}{n}\right)$$

PI includes **both**: uncertainty in estimating the mean **and** natural individual-to-individual spread.

CI vs PI Summary

What	Formula
CI for μ, σ known	$\bar{X}\pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
CI for μ, σ unknown	$\bar{X}\pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$
CI for p	$\hat{p}\pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$
PI, σ unknown	$\bar{X}\pm t_{\alpha/2,n-1}\,S\sqrt{1+\frac{1}{n}}$

CH9: TWO-SAMPLE CIS

$\mu_1-\mu_2$ (known σ_1,σ_2)

$$(\bar{X}_1-\bar{X}_2)\pm z_{\alpha/2}\underbrace{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}_{\text{SE of }(\bar{X}_1-\bar{X}_2)}$$

$\mu_1-\mu_2$ (unknown, equal variances: Pooled t)

Pooled variance estimator:

$$S_p^2=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$$

$$\mathrm{df}=n_1+n_2-2$$

$$(\bar{X}_1-\bar{X}_2)\pm t_{\alpha/2,\,n_1+n_2-2}\,S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$$

$\mu_1-\mu_2$ (unknown, unequal variances: Welch t)

$$(\bar{X}_1-\bar{X}_2)\pm t_{\alpha/2,\,\nu}\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}$$

$$\nu^*\approx\frac{\left(\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1}+\frac{(S_2^2/n_2)^2}{n_2-1}}$$

Use df $\nu=\lfloor\nu^*\rfloor$ (round DOWN to nearest integer)

Paired Data (μ_D)

Define $D_i=X_i-Y_i$ for each pair.

$$\bar{D}=\frac{1}{n}\sum_{i=1}^n D_i,\qquad S_D^2=\frac{1}{n-1}\sum_{i=1}^n(D_i-\bar{D})^2$$

$$\mathrm{Var}(D_i)=\sigma_D^2,\qquad \mathrm{df}=n-1$$

$$\bar{D}\pm t_{\alpha/2,\,n-1}\frac{S_D}{\sqrt{n}}$$

Two Proportions (p_1-p_2)

$$(\hat{p}_1-\hat{p}_2)\pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1}+\frac{\hat{p}_2\hat{q}_2}{n_2}}$$

CH9: VARIANCE CIS

Single Variance (σ^2)

$$X=\frac{(n-1)S^2}{\sigma^2}\sim\chi^2_{n-1}\quad(X\text{ is chi-square})$$

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\,n-1}}\right)$$

Ratio of Variances (σ_1^2/σ_2^2): F Distribution

$$F=\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F_{n_1-1,n_2-1}$$

$$\left(\frac{S_1^2/S_2^2}{F_{\alpha/2,\,n_1-1,\,n_2-1}},\ \frac{S_1^2}{S_2^2}F_{\alpha/2,\,n_2-1,\,n_1-1}\right)$$

CH9: MLE (COMPLETE)

MLE Setup

Likelihood: $L(\theta)=\prod_{i=1}^n f(x_i;\theta)$

Log-likelihood: $\ell(\theta)=\ln L(\theta)$

MLE: solve $\frac{d\ell}{d\theta}=0$ (or score equations for multi-parameter), then check max.

Common MLEs

Bernoulli/Binomial p: $\hat{p}=\bar{X}=x/n$

Normal mean: $\hat{\mu}=\bar{X}$

Normal variance:

$$\hat{\sigma}^2_{\text{MLE}}=\frac{1}{n}\sum_{i=1}^n(X_i-\bar{X})^2$$

MLE for σ^2 uses n (biased); unbiased uses $n-1$.

MLE Properties

Consistency: $\hat{\theta}\to\theta$ as $n\to\infty$.

Asymptotic normality: $\hat{\theta}$ approx normal for large n.

Asymptotic efficiency: achieves minimum possible asymptotic variance.

Invariance: MLE of $g(\theta)$ is $g(\hat{\theta})$.

CH10: HYPOTHESIS TESTING

Core Setup

H_0 (status quo), H_a (claim), choose α, compute statistic, get p-value/critical rule.

Type I error: $\alpha=P(\text{reject }H_0\mid H_0\text{ true})$.

Type II error: $\beta=P(\text{fail reject }H_0\mid H_0\text{ false})$.

Power $=1-\beta$.

Test Statistics (Quick)

One mean, σ known:

$$Z=\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$$

One mean, σ unknown:

$$T=\frac{\bar{X}-\mu_0}{S/\sqrt{n}}\sim t_{n-1}$$

Two means, σ_1,σ_2 known:

$$Z=\frac{(\bar{X}_1-\bar{X}_2)-d_0}{\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}}$$

Two means, unknown equal variances (Pooled t):

$$S_p^2=\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2},\qquad T=\frac{(\bar{X}_1-\bar{X}_2)-d_0}{S_p\sqrt{1/n_1+1/n_2}},\qquad df=n_1+n_2-2$$

Two means, unknown unequal variances (Welch t):

$$T=\frac{(\bar{X}_1-\bar{X}_2)-d_0}{\sqrt{S_1^2/n_1+S_2^2/n_2}},\qquad df=\lfloor\nu^*\rfloor$$

$$\nu^*=\frac{\left(S_1^2/n_1+S_2^2/n_2\right)^2}{(S_1^2/n_1)^2/(n_1-1)+(S_2^2/n_2)^2/(n_2-1)}$$

Paired t-test:

$$D_i=X_i-Y_i,\qquad T=\frac{\bar{D}-d_0}{S_D/\sqrt{n}}\sim t_{n-1}$$

Single variance:

$$\chi^2=\frac{(n-1)S^2}{\sigma_0^2}\sim\chi^2_{n-1}$$

Two variances:

$$F=\frac{S_1^2}{S_2^2}\sim F_{n_1-1,n_2-1}$$

Proportion Tests

One proportion:

$$Z=\frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$$

Two proportions $H_0: p_1=p_2$:

$$\hat{p}=\frac{x_1+x_2}{n_1+n_2},\qquad Z=\frac{\hat{p}_1-\hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\,(1/n_1+1/n_2)}}$$

Goodness-of-Fit

$$\chi^2=\sum_{i=1}^k\frac{(O_i-E_i)^2}{E_i},\qquad df=k-1-m$$

m = number of estimated parameters; typically need each $E_i\geq 5$.

Useful Rules

One-sided right test: reject if statistic $>c_\alpha$.

Two-sided test: reject if $|\text{statistic}|>c_{\alpha/2}$.

p-value rule: reject H_0 if p-value $<\alpha$.

CI-test link: if null value is outside the $(1-\alpha)$ CI, reject corresponding two-sided test.

Sample Size for Target Power (z-test)

For one-sided mean test with effect size $\Delta=|\mu_1-\mu_0|$:

$$n=\left(\frac{(z_\alpha+z_\beta)\sigma}{\Delta}\right)^2$$

For two-sided, replace z_α with $z_{\alpha/2}$.

CH11: SIMPLE LINEAR REGRESSION

Model + Estimators

$$Y_i=\beta_0+\beta_1x_i+\varepsilon_i,\quad \varepsilon_i\overset{iid}{\sim}N(0,\sigma^2)$$

$$S_{xx}=\sum(x_i-\bar{x})^2,\quad S_{yy}=\sum(y_i-\bar{y})^2,\quad S$$

$$b_1=\frac{S_{xy}}{S_{xx}},\quad b_0=\bar{y}-b_1\bar{x},\quad \hat{y}=b_0+b_1x$$

Error + ANOVA Pieces

$$e_i=y_i-\hat{y}_i,\quad \mathrm{SSE}=\sum e_i^2,\quad \mathrm{SSR}=\sum(\hat{y}$$

$$\mathrm{SST}=\mathrm{SSR}+\mathrm{SSE},\qquad s^2=\mathrm{MSE}=\frac{\mathrm{SSE}}{n-2},$$

$$R^2=\frac{\mathrm{SSR}}{\mathrm{SST}}=1-\frac{\mathrm{SSE}}{\mathrm{SST}}$$

Inference on Slope β_1

$$H_0:\beta_1=0,\qquad T=\frac{b_1-0}{s/\sqrt{S_{xx}}}\sim t_{n-2}$$

$$\beta_1\text{ CI: }b_1\pm t_{\alpha/2,n-2}\frac{s}{\sqrt{S_{xx}}}$$

Equivalent ANOVA test:

$$F=\frac{\mathrm{SSR}/1}{\mathrm{SSE}/(n-2)}\sim F_{1,n-2},\qquad F=t^2\ (\text{si}$$

Mean Response CI vs Prediction PI

At x_0:

$$\hat{y}_0=b_0+b_1x_0$$

$$\mu_{Y|x_0}\text{ CI: }\hat{y}_0\pm t_{\alpha/2,n-2}\,s\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}$$

$$Y_0\text{ PI: }\hat{y}_0\pm t_{\alpha/2,n-2}\,s\sqrt{1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}$$

PI is wider than CI (includes individual noise + mean-estimation uncertainty).

Correlation Link

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}, \qquad b_1 = r\frac{S_y}{S_x}$$

Testing $\beta_1 = 0$ is equivalent to testing $\rho = 0$.

Assumptions + Diagnostics

LINE assumptions: linearity, independence, normal errors, equal variance.

Use residual plots (residual vs fitted, QQ) to check assumptions.

Lack-of-Fit (with Replicates)

$$\text{SSLF} = \text{SSE} - \text{PESS}$$

$$F_{\text{LOF}} = \frac{\text{SSLF}/(k-2)}{\text{PESS}/(n-k)} \sim F_{k-2,n-k}$$

k = number of distinct x values.